

No. of Functional Features Included in the Solution

Date	12 August, 2026
Team ID	Individual
Project Name	AI Knowledge Retention Predictor
Maximum Marks	5 Marks

Functional Features Overview:

This document outlines all functional features included in the **AI Knowledge Retention Predictor** solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1.	Ebbinghaus Decay Prediction	Calculates logarithmic memory retention curves based on study dates and difficulty factors.	Decay Engine	Done	High
2.	Dynamic AI Quiz Generation	Leverages Google Gemini API to generate personalized multiple-choice questions per topic.	Gemini Integration	Done	High
3.	Interactive Retention Analytics	Renders dynamic retention decay visual charts and review timelines using Plotly.	Streamlit Dashboard	Done	Medium
4.	Study Session & Score Tracking	Persists user quiz results, review intervals, and retention scores.	SQLite Database	Done	Medium
5.	Optimal Review Scheduler	Predicts exact future dates when user retention drops below 60% threshold.	Scheduling Logic	Done	Medium

6.	User Profile & History Log	Stores user learning history and past quiz logs across sessions.	SQLite / Auth	Done	Low
7.	Custom Theme & Streamlit UI	Responsive web interface featuring dark/light aesthetic modes.	UI Component	Done	Low
8.	Push / Email Reminders	Automated notifications triggered on optimal review dates.	Notification Service	Pending	Low

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1.	User Interface (UI)	2	Interactive Plotly retention curves, responsive Streamlit layout.
2.	Backend / Logic	2	Mathematical Ebbinghaus decay calculation engine, review interval scheduler.

3.	Database / Storage	2	User study session records, persistent quiz history tables in SQLite.
4.	API / Integration	1	Google Gemini API integration for adaptive quiz generation.
5.	Security / Authentication	1	Local session management and API key environment isolation.